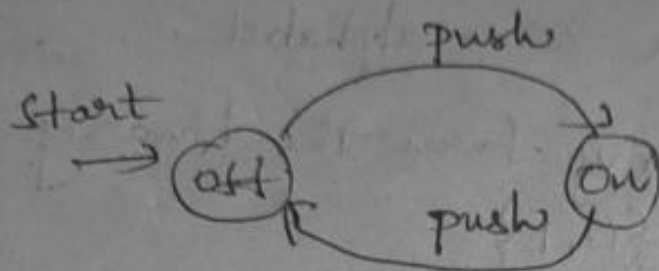


- Automata theory is the study of abstract computing devices or machines
- Formal languages helps us representing the above in a universally recognizable way.

Ex: Consider switch is a m/c

The finite automata is



Where the formal lang is 'push'

Alphabet:

An alphabet is a finite set of symbols represented using the Σ symbol.

→ Alphabets can be binary alphabet which include 0 & 1

→ character alphabet which include the lower case alphabets 'a' to 'z' or anything

$\Sigma = \{0, 1\}$ is a binary alphabet

$\Sigma = \{a, b, c, \dots, z\}$ is a character alphabet

Powers of alphabets:

ex: $\Sigma^0 = \{\epsilon\} \Rightarrow$ only with length '0' (empty string)

$\Sigma^1 = \{a, b, c\}$

$\Sigma^2 = \Sigma^1 \Sigma^1$

$$\Sigma^3 = \Sigma^2 \cdot \Sigma^1$$

$$= \{aa, ab, ac, ba, bb, bc, ca, cb, cc\}.$$

$$\{a, b, c\}$$

$$\Sigma^3 = \{aaa, aab, aac, aba, abb, abc, aca, acb, acc, \dots\}$$

Strings: A string is a finite sequence of symbols chosen from some alphabet.

Ex- 01101 is a string from the binary alphabet $\Sigma = \{0, 1\}$

empty string:

The empty string is the string with 'zero' occurrences of symbols. This string is denoted by ϵ

length of a string is the no. of positions for symbols in the string

ex: 01101 has length 5

The standard notation for the length of a string w is $|w|$

Ex: ① $\rightarrow 011$ length = 3

$$|w| = 3$$

$$\textcircled{2} |\epsilon| = 0$$

Concatenation of strings

Let x & y be strings. Then xy denotes the concatenation of x & y i.e. the string formed by making a copy of x & followed by a copy of y .

If x is the string composed of i symbols $x = a_1 a_2 \dots a_i$ & y is the string composed of symbols $y = b_1 b_2 \dots b_j$ then xy is the string of length $i+j$ &

$$xy = a_1 a_2 \dots a_i b_1 b_2 \dots b_j$$

Ex:- Given two strings $x = 1101$ & $y = 0011$ then

$$xy = 11010011$$

$$yx = 00111101$$

$$\text{length } x+y = 8 \quad \text{or} \quad |x+y| = 8$$

$$x \cdot y \neq y \cdot x \quad (\text{it does not obey commutative})$$

$$x = aa \quad y = ab, \quad z = bb$$

$$\begin{aligned} (x \cdot y) \cdot z &= (aa \cdot ab) \cdot bb \\ &= (aaab) \cdot bb = aaabbb \end{aligned}$$

$$\begin{aligned} x \cdot (y \cdot z) &= aa(ab \cdot bb) \\ &= aa(abbb) = aaabbb \end{aligned}$$

Substring:

if x is a substring of y &
Consider.

~~xy~~
 $x \subseteq y$

~~xy~~ $x \subseteq y$

$$wxz \in y$$

then if w is null then x is prefix
& if z is null then x is suffix.

Language: The set of non-empty strings from alphabet Σ is denoted by Σ^+ . Thus 2 appropriate equivalence are

$$\Sigma^+ = \Sigma^1 \cup \Sigma^2 \cup \Sigma^3 \cup \dots$$

$$\Sigma^* = \Sigma^+ \cup \{\epsilon\}, \quad \Sigma^* = \Sigma^0 \cup \Sigma^1 \cup \Sigma^2 \cup \dots$$

the set of all strings over an alphabet Σ is denoted by Σ^*

Ex- $\{0,1\}^* = \{\epsilon, 0, 1, 00, 01, 10, 11, 000, \dots\}$

$$\{0,1\}^+ = \{0, 1, 00, 01, 10, 11, 000, \dots\}$$

Languages

A set of strings all of which are chosen from some Z^* , where Z is a particular alphabet, is called a language.

If Z is an alphabet, & $L \subseteq Z^*$, then L is a language over Z .

Ex- $Z = \{0, 1\}$ $L = \{00, 01\}$

$$Z^* = \{\epsilon, 0, 1, 00, 01, 10, \dots\}$$

$$L^* = \{\epsilon, 00, 01, 0000, 0001, \dots, 000000, 000001, \dots\}$$

examples of languages

- ① The language of all strings consisting of n 0's followed by n 1's, for some $n \geq 0$

$$\{\epsilon, 01, 0011, 000111, \dots\}$$

(n 0's followed by n 1's)

- ② The set of strings of 0's & 1's with an equal number of each.

$$\{\epsilon, 01, 10, 0011, 0101, 1001, \dots\}$$

3) The set of binary numbers whose Value is a prime

$$\{10, 11, 101, 111, 1011, \dots\}$$

$$2 - 10$$

$$3 - 11$$

$$5 - 101$$

$$7 - 111$$

$$11 - 1011$$

$$13 -$$

$$17 -$$

4) Σ^* is a language for any alphabet Σ

5) ϕ , the empty language, is a language over any alphabet.

6) $\{\epsilon\}$, the lang containing ϵ - empty string
 ~~$\{\epsilon\}$ is empty lang~~
 but $\phi \neq \{\epsilon\}$

$\phi \rightarrow$ has no strings. & $\{\epsilon\}$ has one string that is ϵ .

operations

(1) Union of sets (U)

If A & B are the given sets, then the resultant set C on taking Union of A & B may be represented as

$$C = \{x \mid x \in A \text{ or } x \in B\}$$

② Intersection ($\cap$)

$$C = \{x \mid x \in A \text{ \& } x \in B\}$$

③ Disjoint Sets

Two sets are said to be disjoint when intersection of the 2 given sets results in a null set.

$$C = \{x \mid x \notin A \text{ and } x \notin B\}$$

$$A = \{1, 2\} \text{ \& } B = \{3, 4\}$$

$$A \cap B = \{\}$$

$$= \phi$$

A \& B are disjoint sets

④ Set Difference:

Set difference is a binary operation performed on two sets \& results in a set of elements that are present in the first set \& not present in the second.

$$= \{1, 2, 3\} - \{2, 3, 5\}$$

$$= \{1\}$$

⑤ Complement of a set (S')

Consider S is a set

$$\text{then } S' = U - S$$

Complement of a set can be defined as the set difference of the universal set \& the given set.

⑥ Symmetric Difference ④

Symmetric difference of 2 sets is a set of elements present in any of the given sets

$$\begin{aligned} A \oplus B &= (B - A) \cup (A - B) \\ &= (A \cup B) - (A \cap B) \end{aligned}$$

⑦ Cartesian product (X):

Cartesian product of the 2 sets A & B is a set of all possible ordered pairs (a, b) where $a \in A$ & $b \in B$

$$\begin{aligned} A \times B &= \{J, M\} \times \{x_1, x_2, x_3\} \\ &= \{(J, x_1), (J, x_2), (J, x_3), (M, x_1), (M, x_2), (M, x_3)\} \end{aligned}$$

Finite State Machine:

The finite state system ^{is} ~~represents~~ a mathematical model of a system with certain i/p. The model finally gives certain o/p

→ The i/p when given to the machine, the i/p is processed by various states of the m/c. These states are called intermediate states.

→ The finite state system is a very good design tool for the programs such as text editors & lexical analyzers.

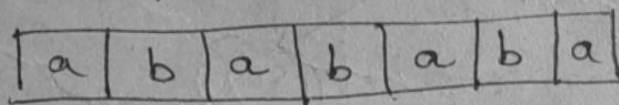
Definition:

A finite automata is a collection of 5-tuples $(Q, \Sigma, \delta, q_0, f)$ where

- Q is a finite set of states, which is non-empty
- Σ is the i/p alphabet (i/p set)
- q_0 is an initial state & q_0 is in Q i.e., $q_0 \in Q$
- f is a set of final states.
- δ is a transition function or a mapping function using this function the next state can be determined.

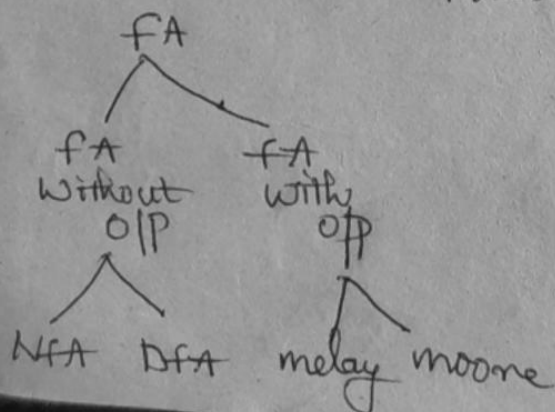
Finite Automata Model

i/p tape

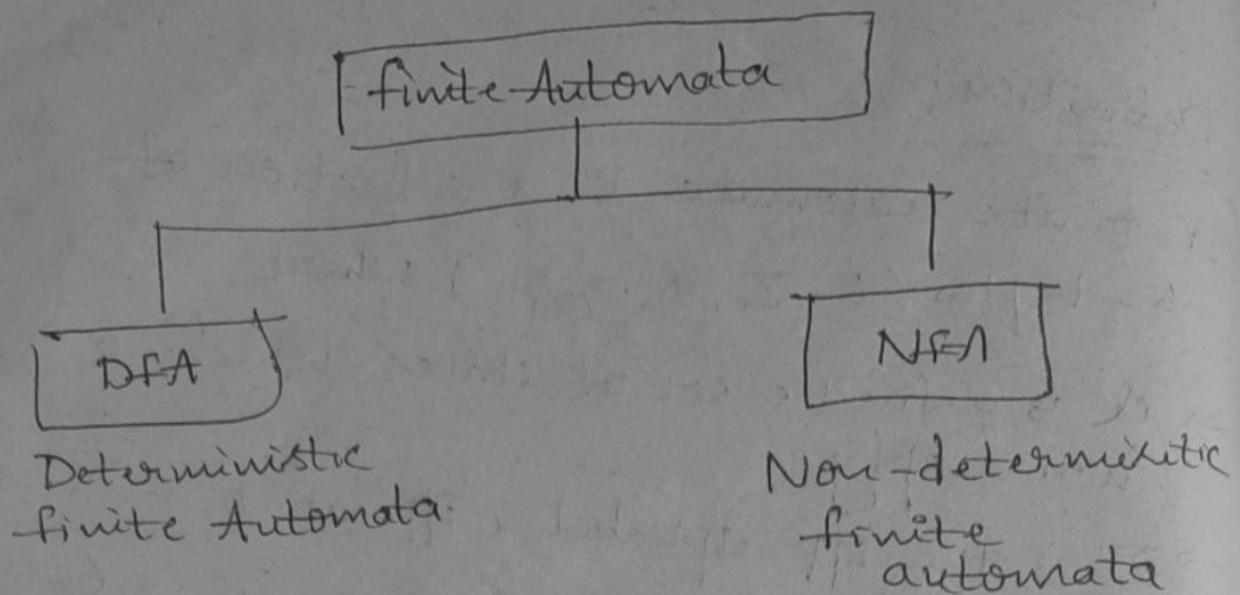


↑ Tape reader reading the i/p symbol

Finite Control



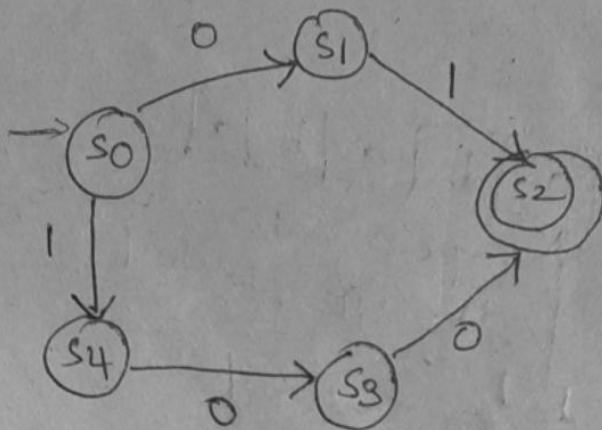
Types of Automata



Acceptance of strings & Languages

The strings ~~& lang~~ can be accepted by a finite automata, when it reaches to a final state.

① Transition diagram

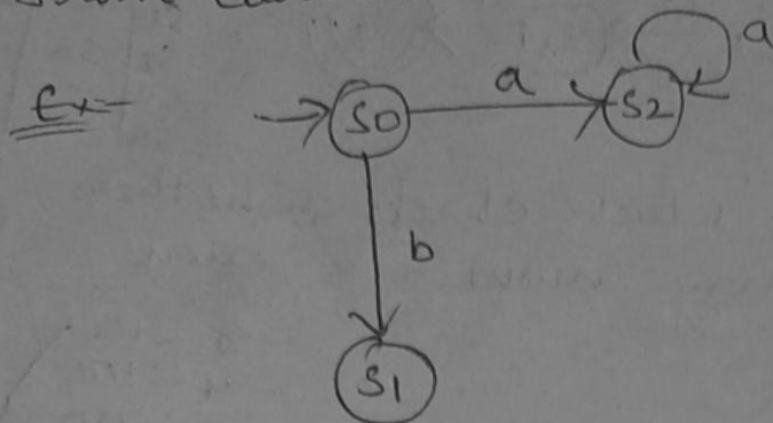


Transition table

States \ IP	0	1
S0	S1	S4
S1		S2
S2	S2	
S3	S3	
S4		

Deterministic finite Automaton

There is only one path for a specific i/p from current state to next state.



→ The DFA can be represented by the same 5-tuples described in the definition of FSM.

* The finite set of states which can be denoted by Q .

* The finite set of i/p symbol Σ

* The start state q_0 such that $q_0 \in Q$

* A set of final states f such that $f \subseteq Q$

* The mapping function or transition function denoted by δ .

→ The DFA is a five tuple notation denoted as

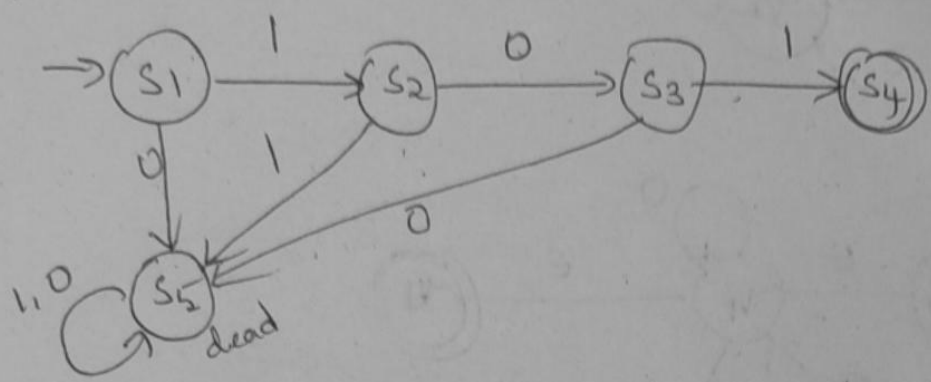
$$A = (Q, \Sigma, \delta, q_0, f)$$

DFA

1) Design FA that accepts string 101

$$\Sigma = \{0, 1\}$$

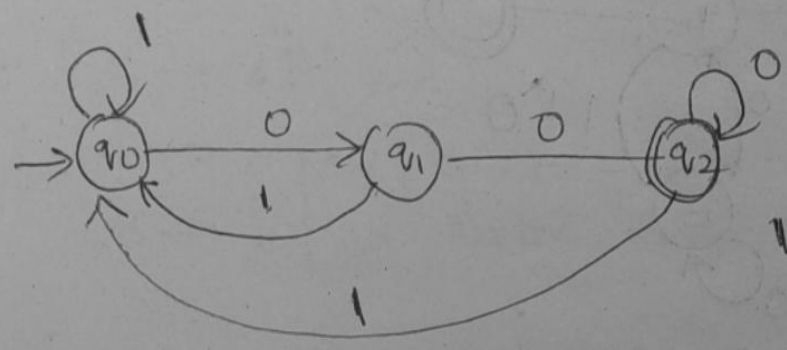
$$L = \{101\}$$



101 ✓
1001 ✗
1110

2) Design FA to accept the strings that always ends with 00

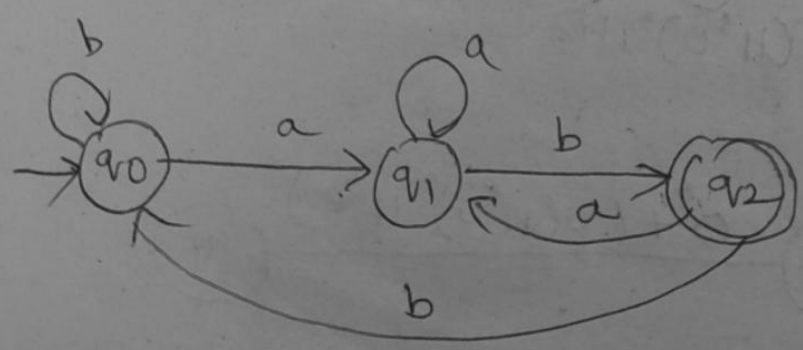
$$L = \{00, 100, 000, 1100, \dots\}$$



00
000
100
1000
101100100
10111100
10100

3) DFA that ends with ab

$$L = \{ab, aab, bab, \dots\}$$

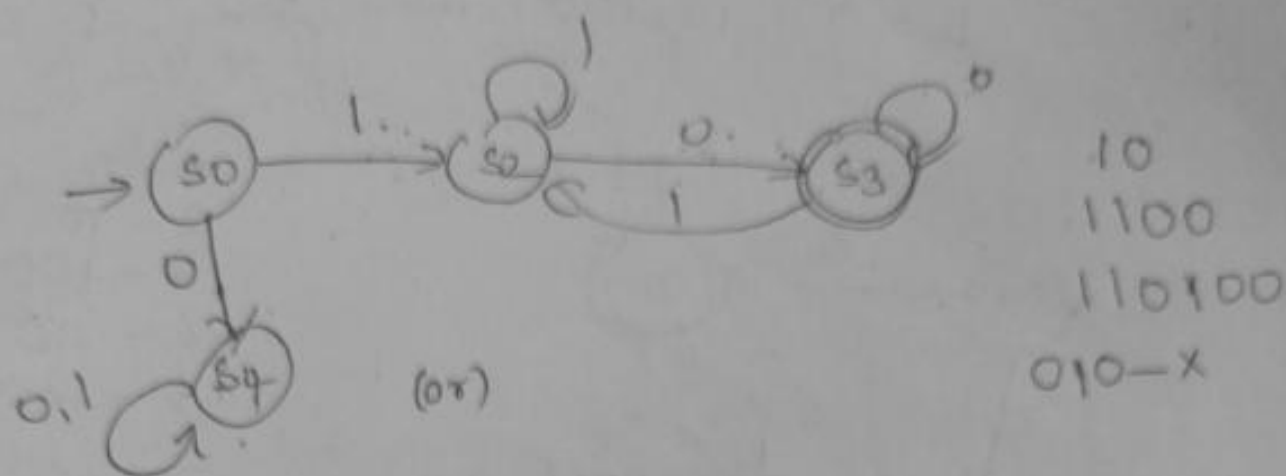


aabbab ✓
abbaab

bab
ab

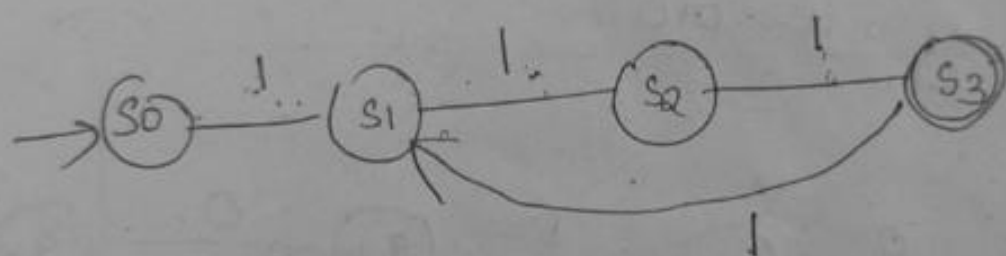
4) Design FA which accepts only those strings which start with 1 & ends with 0.

$$L = \{10, \cancel{110}, 110 - \}$$



5) Design FA that checks whether the given binary no. is divisible by 3

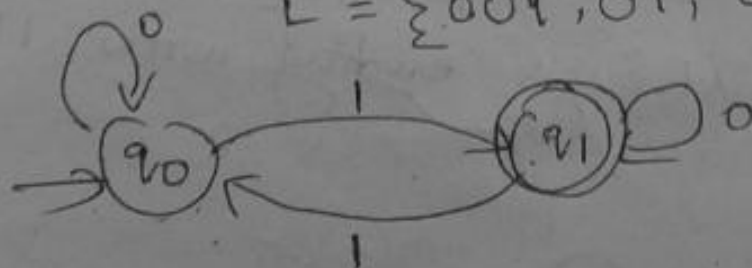
$$L = \{111, 11111, 1111111 - - - \}$$



6) Design FA which accepts odd no. of 1's & any no. of 0's

$$\Sigma = \{0, 1\}$$

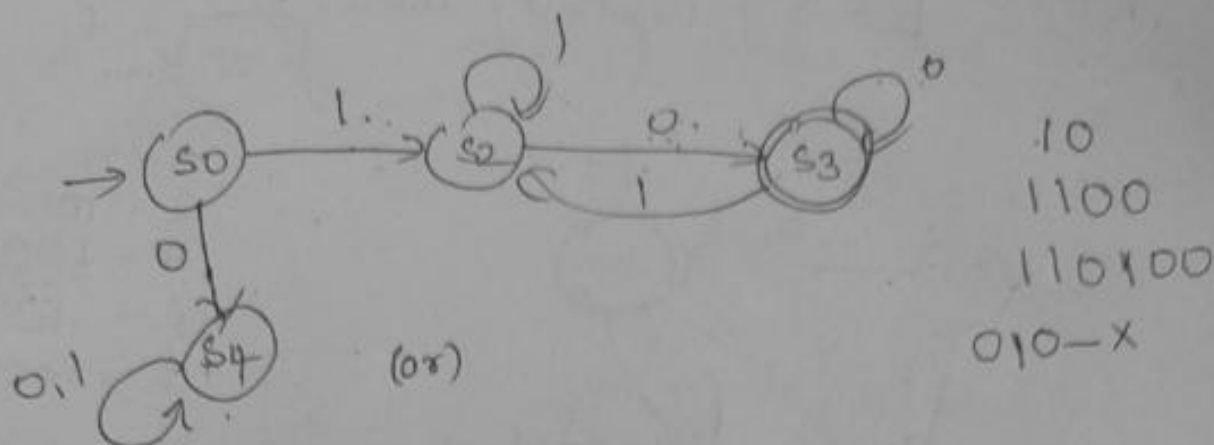
$$L = \{001, 01, 010101 - - \}$$



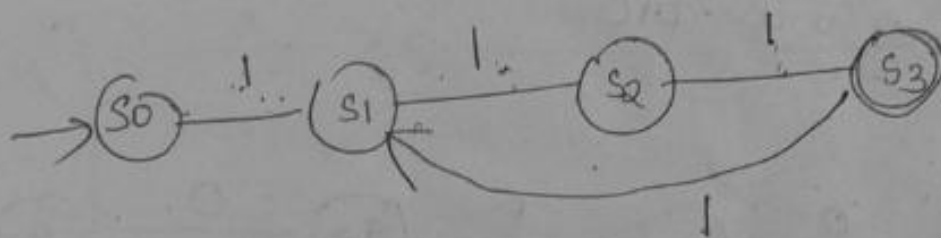
01
00111

- 4) Design FA which accepts only strings which start with 1 & ends with 0.

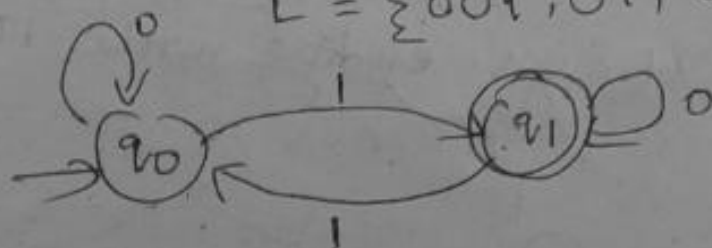
$$L = \{10, \cancel{010}, 110, \dots\}$$



- 5) Design FA that checks whether the given unary no. is divisible by 3
- $$L = \{ \text{|||}, \text{|||||}, \text{|||||||}, \dots \}$$



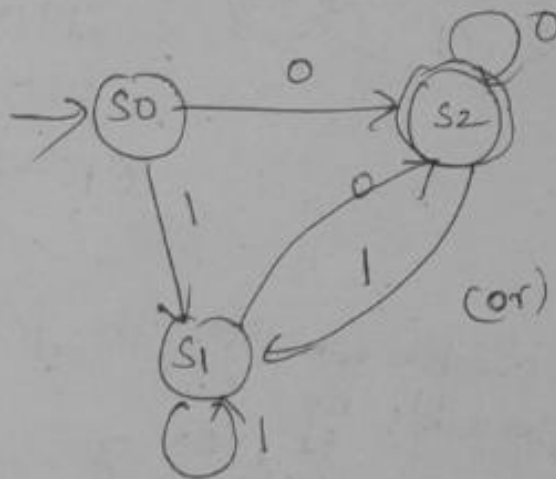
- 6) Design FA which accepts odd no. of 1's & any no. of 0's
- $$\Sigma = \{0, 1\}$$
- $$L = \{001, 01, 010101, \dots\}$$



01
00111

- ⑦ Design FA which checks whether the given binary no. is even over $Z = \{0, 1\}$

$$L = \{0, 10, 100, 110, 1000, \dots\}$$

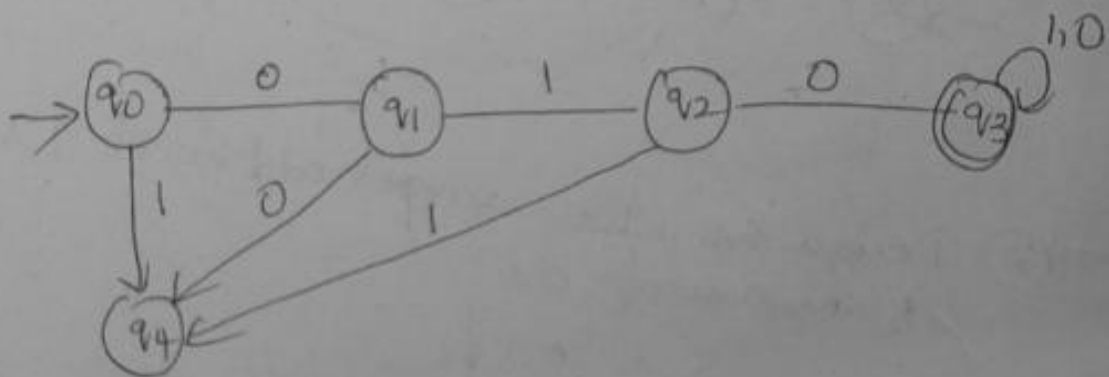


0 - 0
2 - 10
4 - 100
6 - 110
8 - 1000

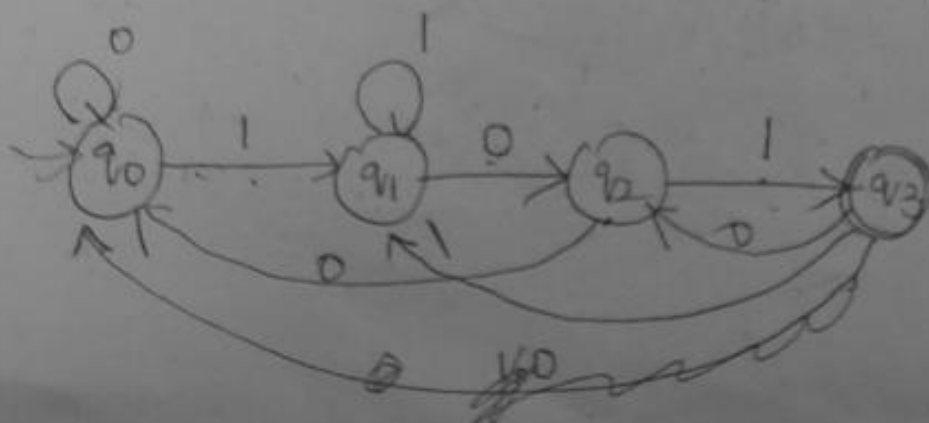
(or)



- ⑧ Design FA that accepts set of strings beginning with 010



- ⑨ FA for strings ending with 101

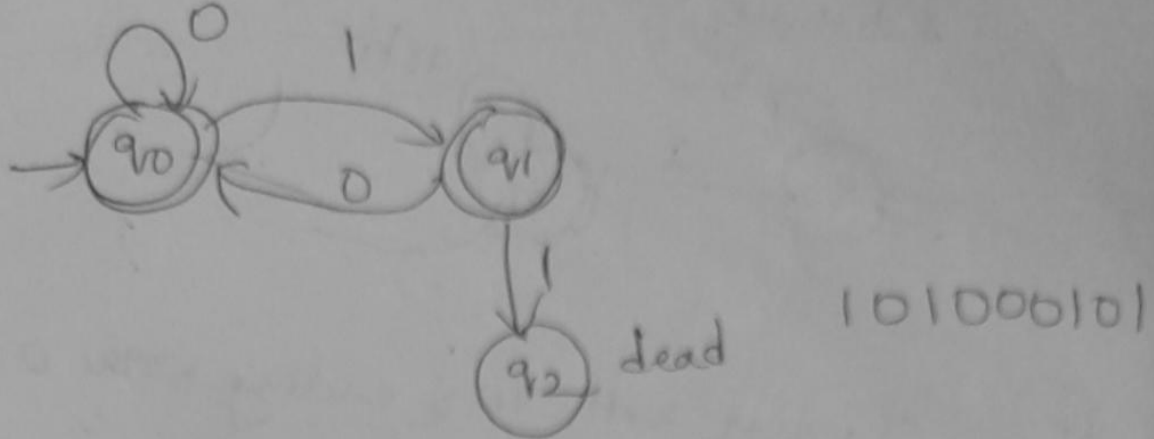


1010101
1010101

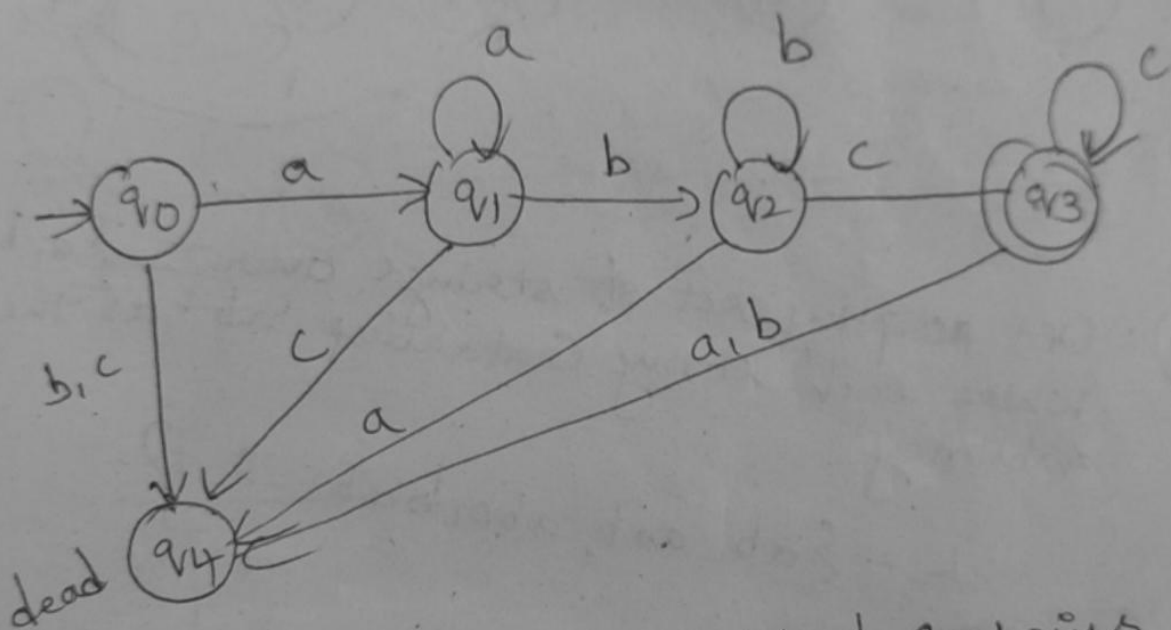
10110101

10011101
01101

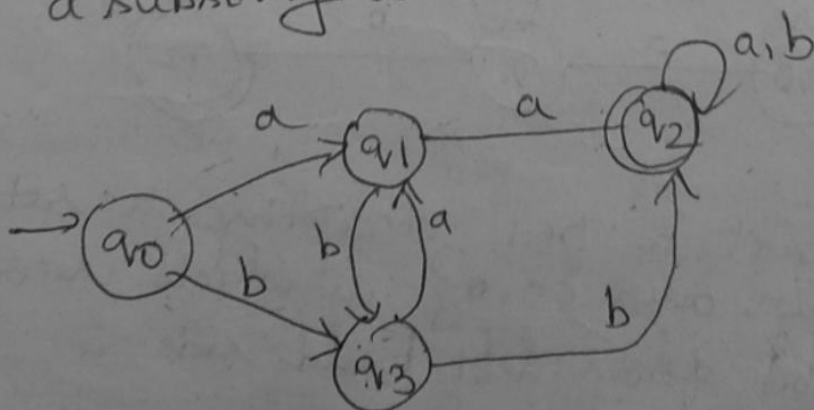
- (10) w is a string which doesn't contain consecutive 1's



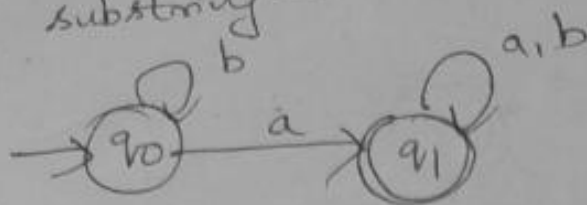
- (11) FA for $L = \{a^n b^m c^l \mid n, m, l \geq 1\}$
 $L = \{abc, aabc, abbc, \dots\}$



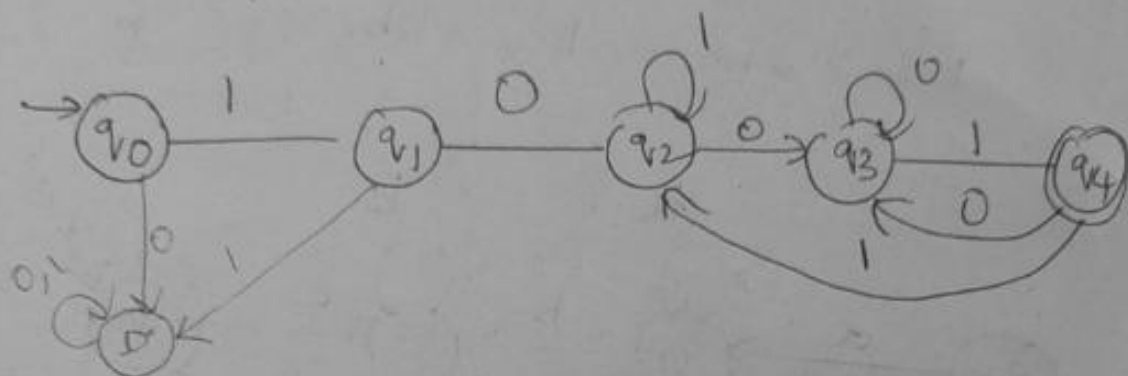
- (12) DFA, every string accepted contains a substring aa or bb



- (13) DFA, every string contains 'a' as substring

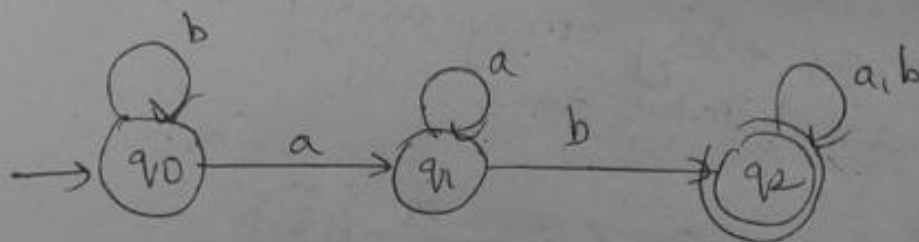


- (14) Starting with 10 & ending with 01

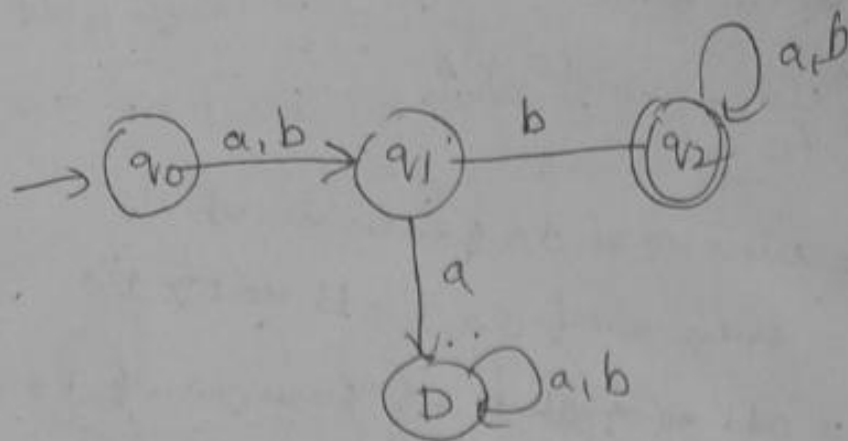


- (15) DFA accepting set of strings over $\Sigma = \{a, b\}$ where each string contains 'ab' as the substring

$$L = \{ab, aab, abb, bab, \dots\}$$



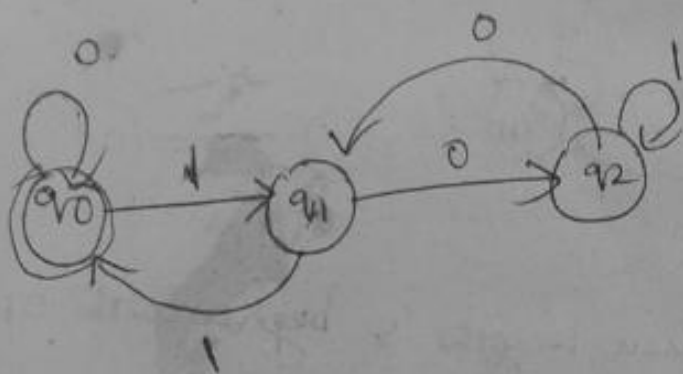
- (16) Construct a DFA accepting a set of strings over $\{a, b\}$ in which second symbol from left hand side is always 'b'.



17. Design FA to check whether a given no. is divisible by 3. over $\Sigma = \{0, 1\}$

✓ $L = \{0, 3, 6, 9, \dots\}$

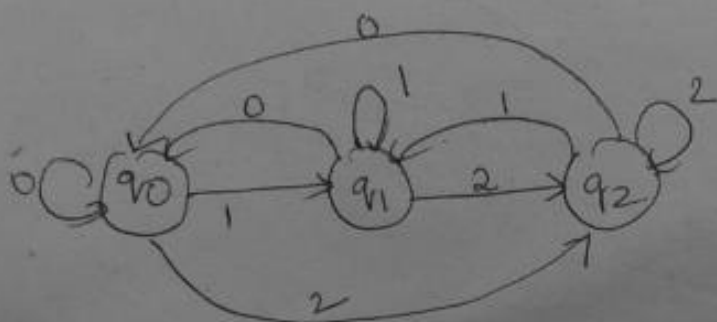
$0 \bmod 3 = 0 \rightarrow q_0$
 $01 \rightarrow 1 \bmod 3 = 1 \rightarrow q_1$
 $10 \rightarrow 2 \bmod 3 = 2 \rightarrow q_2$
 $100 \rightarrow 3 \bmod 3 = 0 \rightarrow q_0$
 $1000 \rightarrow 4 \bmod 3 = 1 \rightarrow q_1$
 $101 \rightarrow 5 \bmod 3 = 2 \rightarrow q_2$



	0	1
q_0	q_0	q_1
q_1	q_2	q_0
q_2	q_1	q_2

$0 \bmod 3 = 0 \rightarrow q_0$
 $1 \bmod 3 = 1 \rightarrow q_1$
 $2 \bmod 3 = 2 \rightarrow q_2$

Same for $\Sigma = \{0, 1, 2\}$



	0	1	2
q_0	q_0	q_1	q_2
q_1	q_2	q_0	q_1
q_2	q_1	q_2	q_0

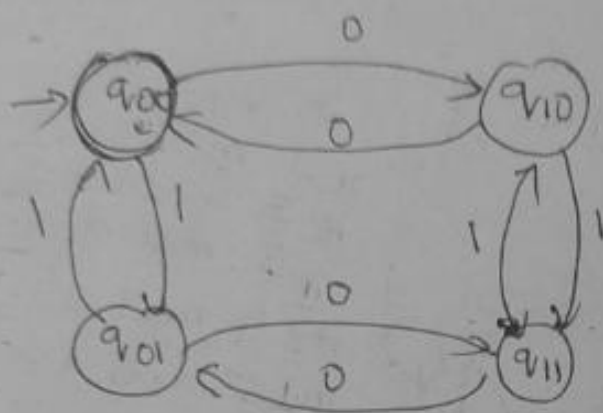
18) Design FA which accepts even no. of 0's & even no. of 1's

q_{00} = even no. of 0's & even no. of 1's

q_{01} = even no. of 0's & odd no. of 1's

q_{10} = odd no. of 0's & even no. of 1's

q_{11} = odd no. of 0's & odd no. of 1's



19) $L = \{ w \mid w \text{ is even length \& begins with } 01 \}$

